

MIAE 215 · Mini-Course Lesson 3

Variable Types, Memory Allocation & Keyboard Input

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Source: [Lesson 3 Web Page](#) | Video: [lesson3_variable_types.mp4](#)

Table of Contents

- 1. [Core Primitive Data Types](#)
- 2. [Memory Sizing & sizeof\(\) Inspection](#)
- 3. [Keyboard Stream Input: cin >>](#)
- 4. [Integer Division vs. Floating Division](#)
- 5. [Teacher Code & Exercise Walkthroughs](#)

1. Core Primitive Data Types

In Lesson 3, Prof. Gordon introduces the five fundamental data types used in engineering computation:

Type	Size (Bytes)	Range / Precision	Typical Engineering Purpose
int	4 bytes	-2.14×10^9 to $+2.14 \times 10^9$	Integer counters, loop indices, discrete states.
float	4 bytes	$\pm 10^{-38}$ to $\pm 10^{+38}$ (~7 digits)	Embedded memory buffers, low-precision signals.
double	8 bytes	$\pm 10^{-308}$ to $\pm 10^{+308}$ (~16 digits)	Standard for physics, aerodynamics, and robotics.
char	1 byte	-128 to +127 (ASCII code)	Keyboard keys, ASCII text, single telemetry bytes.
bool	1 byte	true (1) or false (0)	Decision flags, binary sensor triggers.

2. Memory Sizing & sizeof() Inspection

C++ provides the compile-time operator `sizeof()` to measure the exact byte footprint of any type or variable:

```
int a = 10;
cout << "Size of integer variable a: " << sizeof(a) << " bytes
";
cout << "Size of double type:          " << sizeof(double) << " bytes
";
```

3. Keyboard Stream Input: `cin >>`

To read data entered by the user at the keyboard, C++ uses the standard input stream `cin` and the **stream extraction operator** (`>>`):

```
int i;
double x;

cout << "Enter an integer and a double: ";
cin >> i >> x; // Skips leading whitespace and reads values sequentially
```

Teacher Exam Warning / Pitfall:

When a user enters numeric data with `cin >>` and presses Enter, a newline character (`\n`) remains in the keyboard buffer. If you follow `cin >>` with `getchar()`, the `getchar()` will instantly read the leftover newline without pausing! Call `getchar()` twice to guarantee a pause.

4. Integer Division vs. Floating Division

When dividing two integers, C++ forcibly truncates the result to an integer:

```
int a = 7, b = 2;
cout << a / b;    // PRINTS 3, NOT 3.5!

double x = 7 / 2; // Still evaluates to 3, then converts to 3.0!
double y = 7.0 / 2; // Evaluates to 3.5 (floating-point division)
```

5. Teacher Code & Exercise Walkthroughs

The following source files are provided in this lesson folder:

- [lesson3.cpp](#) : Main lecture code demonstrating variable declarations, `sizeof()`, and `cin`.
- [lesson3_exercises.cpp](#) : Practice questions on variable assignment.
- [lesson3_exercises_solutions.cpp](#) : Fully annotated solutions.
- [lesson3_more.cpp](#) & [lesson3_more_solutions.cpp](#) : Supplemental challenges.